

Interrupt Flag Bits

Concept

Interrupt Flag (*IF*)

- *IF* = 1: Allows the INTR pin to cause an interrupt.
- *IF* = 0: Prevents the INTR pin from causing an interrupt.
- **Setup:** *IF* is set by the STI instruction and cleared by the CLI instruction.

Concept

Trap Flag (*TF*)

- *TF* = 1: Causes an **Interrupt Type 1** to occur after each instruction executes (Single-stepping).
- *TF* = 0: Normal program execution occurs.
- **Setup:** There are **no special instructions** to set or clear *TF* directly. It must be manipulated via the stack.



1. Controlling the TRAP Flag (TF)

The Trap Flag (*TF*) is used for single-stepping through a program. Since there are no direct instructions to set or clear *TF* (unlike *IF* which has STI/CLI), it must be manipulated via the stack.

Example / Procedure

TRON Procedure (Enable Trapping)

This procedure sets the Trap Flag bit (*TF* = 1) to enable single-step mode.

```
TRON PROC
    PUSHF                ; Push current FLAGS register onto stack
    POP    AX             ; Pop FLAGS into AX to modify bits
    OR     AH, 01H        ; Set Trap Flag bit (Bit 8 of FLAGS)
    PUSH  AX              ; Push modified FLAGS back onto stack
    POPF                ; Pop back into FLAGS register
    IRET                 ; Return from procedure
TRON  ENDP
```

Example / Procedure

TROFF Procedure (Disable Trapping)

This procedure clears the Trap Flag bit ($TF = 0$) to disable single-step mode.

```
TROFF PROC
    PUSHF                ; Push current FLAGS register onto stack
    POP    AX            ; Pop FLAGS into AX to modify bits
    AND    AH, 0FEH      ; Clear Trap Flag bit (Bit 8 = 0)
    PUSH    AX           ; Push modified FLAGS back onto stack
    POPF                ; Restore modified FLAGS register
    IRET                ; Return from procedure
TROFF ENDP
```

Rules / Notes

Logic Breakdown

- **Bit Position:** The Trap Flag is at **Bit 8** of the FLAGS register (which is the lowest bit of the high byte *AH*).
- **Masks:** `OR AH, 1` sets the bit, while `AND AH, 0FEH` (11111110) clears it.